

Physics, 2014 EC, natural science, entrance exam, round 1

Grade 12 · Physics · 2014 E.C.

Time allowed: 120 minutes

Questions: 25

Total marks: 25

Name _____ Grade _____ ID No. _____ Date _____

1. An object is placed 25.0 cm in front of a concave mirror having focal length of 10.0 cm. The magnification of the mirror is _____. (1)
 - A. 2.5
 - B. 16.7
 - C. 0.67
 - D. 1.5
2. The unit vector in the direction of $\vec{A} = 3\hat{i} + 4\hat{j}$ is _____. (1)
 - A. $3/5\hat{i} + 4/5\hat{j}$
 - B. $2\hat{i} - \hat{j}$
 - C. $3/2\hat{i} + 4/2\hat{j}$
 - D. $\hat{i} + \hat{j}$
3. Which of the following pair of vectors are collinear? (1)
 - A. $2\hat{i} - 3\hat{j}$ and $-4\hat{i} - 6\hat{j}$
 - B. $\hat{i} + \hat{j}$ and $-\hat{i} + \hat{j}$
 - C. $2\hat{i} + 3\hat{j}$ and $-4\hat{i} + 6\hat{j}$
 - D. $2\hat{i} + 3\hat{j}$ and $-4\hat{i} - 6\hat{j}$
4. Which of the following statements distinguishes between random and systematic errors? (1)
 - A. Systematic error can be minimized by taking several readings and averaging; whereas random error cannot be minimized.
 - B. Systematic error makes measured values to be always above or always below the true value; whereas random error causes measured values to be sometimes above and sometimes below the true value.
 - C. Random error makes measured values to be always above or always below the true value; whereas systematic error causes measured values to be sometimes above and sometimes below the true value.
 - D. Random error is biased, whereas systematic error does not show bias.
5. For an object moving in straight line, (1)
 - A. if the displacement changes by equal amount in equal time interval, the displacement-time graph is horizontal line.
 - B. the area of the region under the displacement-time graph and the slope of velocity-time graph are, respectively, the acceleration and displacement of the object.
 - C. if the velocity increases by equal amount in equal time interval, the velocity-time graph is straight line with positive slope and the acceleration-time graph is horizontal.
 - D. the slope of displacement-time graph and the area of the region under acceleration-time graph are, respectively, the acceleration and velocity of the object.
6. Which of the following is true about satellite motion? (1)
 - A. A satellite moving closer to the Earth takes longer time to complete one revolution than a satellite farther from the Earth.
 - B. For two satellites moving around the Earth at equal orbital radius, a heavier satellite has larger speed than a lighter one.
 - C. For equal orbital radius, a satellite moving around the Earth has smaller speed than a satellite moving about Moon.
 - D. A satellite closer to Earth is faster than a satellite farther from the Earth.

- 7.** A ball of mass m moving with a speed v strikes a hard wall at an angle θ with the wall. It is reflected with the same speed at the same angle as shown in the figure below. If the ball is in contact with the wall for time t , then the average force acting on the wall is given by (1)

 - $mv/t \cos \theta$
 - $mv/t \sin \theta$
 - $2mv/t \cos \theta$
 - $2mv/t \sin \theta$

8. According to the work-energy theorem, (1)

 - work done by tension on an object tied to a string and moving in a horizontal circle is negative.
 - work done by gravity on an object thrown downward is negative.
 - work done by gravity on an object thrown upward and moving upward is negative.
 - work done by gravitational force on a satellite moving around the earth in a circular orbit is positive.

9. Standing wave is formed when two identical waves traveling in opposite directions in the same medium interfere with each other. If y_1 and y_2 are the interfering waves and $y=y_1+y_2$ is the resulting standing wave, which of the following is a correct combination? (1)

 - $y_1=A \sin(kx-\omega t), y_2=A \sin(kx+\omega t), y=A \sin(kx)\cos(2\omega t)$
 - $y_1=A \sin(kx-\omega t), y_2=A \sin(kx+\omega t), y=2A \sin(kx)\cos(\omega t)$
 - $y_1=A \sin(kx-\omega t), y_2=A \sin(kx-\omega t), y=2A \sin(kx)\cos(\omega t)$
 - $y_1=A \sin(kx-\omega t), y_2=A \cos(kx+\omega t), y=2A \sin(kx-\omega t)$

10. A Young double slit experiment consists of two narrow slits separated by 0.06 mm and 1.2 m away from a screen. If the slits are illuminated with a light of wavelength 563 nm , the location of the fourth bright fringe on the screen is (1)

 - 0.394 cm
 - 0.45 cm
 - 3.94 cm
 - 4.5 cm

11. An electric dipole consists of a negative charge $-q$ placed at a point with x and y coordinates $(-2,-2)$ and a positive charge $+q$ at $(2,2)$. The dipole is placed in a uniform electric field E directed in the positive x direction. If the dipole axis makes an angle θ with the field, which of the following is true about the dipole? (1)

 - The field rotates the dipole in the direction of decreasing the angle θ , until the dipole becomes parallel to the field.
 - The field rotates it in the direction of increasing the angle θ , until the dipole is perpendicular to the field.
 - It is in both translational and rotational equilibrium.
 - It travels in the direction of the electric field.

12. The electric potential at a distance of 2 cm from a $-1.6 \times 10^{-9} \text{ C}$ point charge is (1)

 - -14.4 V .
 - 720 V .
 - 4.4 V .
 - -720 V .

13. Two resistors $R_1=1 \Omega$ and $R_2=4 \Omega$ are connected in parallel to each other and their combination is connected in series with a third resistor $R_3=2 \Omega$. The whole combination is connected to a voltage source of potential difference 14 V . The potential difference across R_1 is (1)

 - 10 V .
 - 14 V .
 - 5 V .
 - 4 V .

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- 14.** Consider a long and tightly wound solenoid carrying current I , having length L and N turns of winding. If n is the number of turns per unit length, which of the following describes the magnetic field produced by the solenoid? (1)
- A. The magnetic field inside the solenoid is zero, whereas the field outside is proportional to the length of the solenoid.
 - B. The magnetic field inside the solenoid is constant and is parallel to the solenoid's axis, whereas the field outside is negligible.
 - C. The magnetic field inside the solenoid is uniform and perpendicular to the axis of the solenoid.
 - D. The magnetic field outside the solenoid is as strong as the field inside it.
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- 15.** According to the law of electromagnetic induction, (1)
- A. the magnitude of emf induced in a circuit is directly proportional to the rate of change the magnetic flux crossing any surface bounded by the circuit.
 - B. a changing magnetic field parallel to the plane of a circuit induces an emf in it.
 - C. a constant magnetic flux crossing the surface of a circuit induced an emf in the circuit.
 - D. an emf is induced in a circuit when a constant magnetic field crosses its surface.
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- 16.** According to the Heisenberg's uncertainty principle, (1)
- A. the larger the position uncertainty, the larger the momentum uncertainty.
 - B. if the uncertainty in position decreases, the momentum uncertainty decreases, too.
 - C. the more precisely we measure the position of a particle, we become less precise of its momentum and vice versa.
 - D. we can measure both the position and momentum of a particle simultaneously with unlimited precision.
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- 17.** Which of the following statements is NOT correct about reflection and refraction of sound waves and their applications? (1)
- A. In pre-natal scanning, reflection and refraction of sound allows doctors to monitor the progress of a developing baby.
 - B. Reflection of sound (echo) is formed when sound enters a medium of varying densities.
 - C. Reflection and refraction of sound are used to detect flaws in metals and to help people park their cars.
 - D. Refraction of sound is the change in its direction as it travels through a medium having different densities.
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- 18.** A torque T rotates an object of mass m and moment of inertia I through an angle θ . The work done on the object by the torque is (1)
- | | |
|--------------|--------------|
| A. $I\theta$ | B. $m\theta$ |
| C. $T\theta$ | D. TI |
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- 19.** One of the following statements is a correct description of the charging processes? (1)
- A. During charging a neutral object by induction using a negatively charged rod, charge flows from the rod to the object.
 - B. When a neutral metallic sphere is charged by induction using a negatively charged rod, the sphere acquires a net negative charge.
 - C. When a positively charged object is momentarily brought into contact with a neutral metallic sphere, the sphere acquires a net positive charge.
 - D. When a plastic rod is rubbed with wool, the two objects acquire a net charge of the same sign.

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- 20.** A parallel plate capacitor is charged with a battery. The battery is then disconnected, and a dielectric material is inserted between the plates of the capacitor. Which of the following is correct about the effect of the dielectric? (1)
- A. The potential difference between the plates remains the same and the charge increases.
 - B. The charge on the plates decreases and the energy stored increases.
 - C. The electric field between the plates and the energy stored decrease.
 - D. The capacitance remains the same and the electric field between the plates increases.
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- 21.** Which one of the following statements is correct about the resistance of a conductor with a circular cross-section? (1)
- A. At a given temperature, the resistance of copper and silver wires of the same length and diameter is the same.
 - B. The resistance of the conductor increases as its length increases and its diameter decreases.
 - C. The resistance of the conductor increases as its length and diameter increases.
 - D. The resistance of the conductor decreases as it is heated.
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- 22.** The primary coil of a transformer has 250 turns and its secondary coil consists of 1500 turns. If 220 V is applied to its primary coil, the voltage induced in the secondary coil and the ratio of current in the primary coil to the current in the secondary coil, respectively, are (1)
- A. 1320 V and 6
 - B. 1330 V and 6
 - C. 1220 V and 60
 - D. 1320 V and 60
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- 23.** A diode is an electronic component that (1)
- A. is useful to convert a DC voltage to AC voltage.
 - B. allows electric current to pass through it in one direction.
 - C. allows the flow of current when its p-type region is connected to the negative terminal of a cell.
 - D. allows current to pass through it either from cathode to anode or from anode to cathode.
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- 24.** Three forces, each having equal magnitudes of 10 N, are applied to an object on a horizontal surface. A force F_1 makes an angle of 30° above positive x-axis, F_2 makes 30° above the negative x-axis, and the third force F_3 is directed along the positive x-axis. The magnitude and direction of their resultant force are, respectively (1)
- A. 20 N 45° above positive x-axis.
 - B. 14.1 N in the positive y-axis.
 - C. 20 N in the negative y-axis.
 - D. 14.1 N 45° above positive x-axis.
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- 25.** A student throws a ball vertically upward with a speed of 20 m/s. The maximum height reached by the ball and its velocity 3.0 s after it was thrown are, respectively (1)
- A. 20 m and 30 m/s upward.
 - B. 15 m and 10 m/s, downward.
 - C. 105 m and 50 m/s upward.
 - D. 20 m and 10 m/s downward.
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— END OF EXAMINATION —